Support Vector Machine

Hard Margin SVM-Optimization Problem

- If the data points are less, then it is easy to find the support vectors
- In practical, for large data points, it is not feasible.
- So SVM construction is reformulated into optimization problem
- Any problem whose objective function needs to be maximized or minimized subject to the constraints is called optimization problem.
- Since SVM involves maximization of margin, it can be modelled as optimization problem.

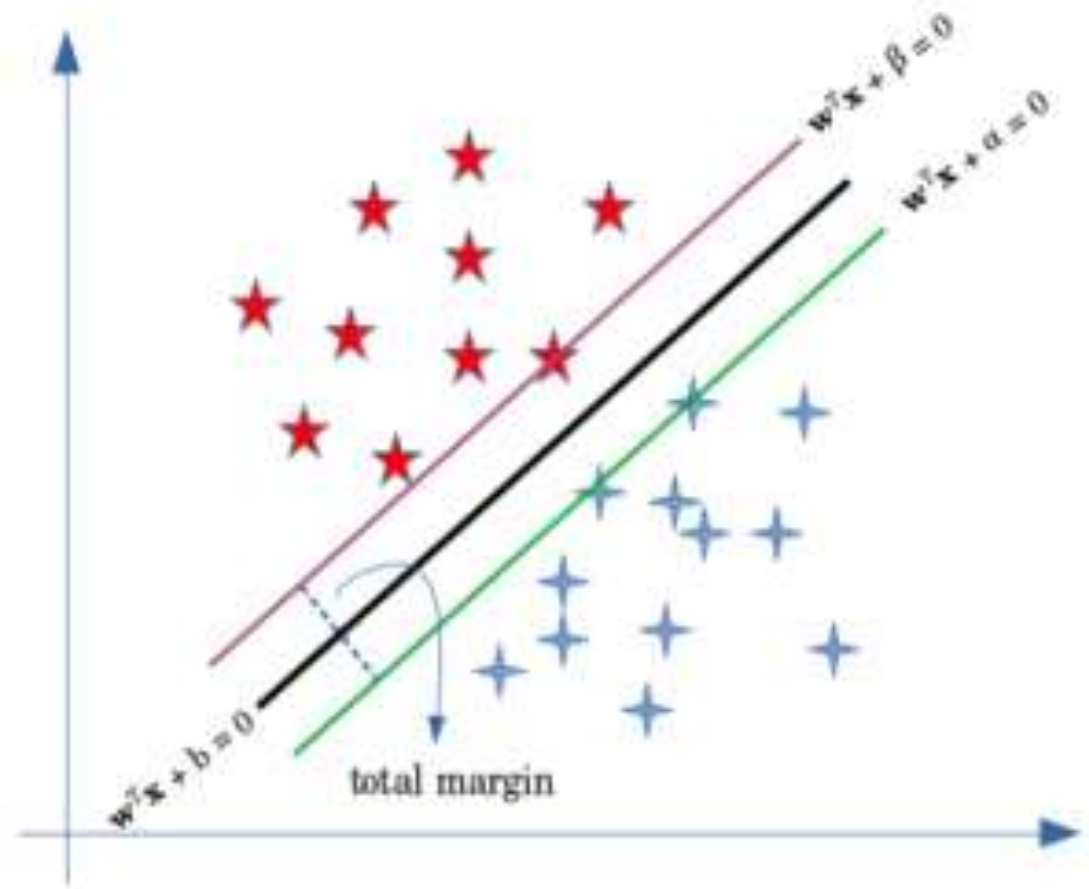
Method for optimal problem

- State the problem as a primal optimization problem
- Convert that to dual Lagrangian problem. Finally using the solutions construct the hyperplanes

Hard Margin Vs Soft margin

- When the data is linearly separable, and we don’t want to have any misclassifications, we use SVM with a hard margin.
- However, when a linear boundary is not feasible, or we want to allow some misclassifications in the hope of achieving better generality, we can opt for a soft margin for our classifier.

SVM with a Hard Margin



- we can define the margin by two parallel hyperplanes:

$$w^T x + \alpha = 0,$$

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- Without allowing any misclassifications in the hard margin SVM, we want to maximize the distance between the two hyperplanes.

- We want to maximize this margin. Without the loss of generality, we can consider $\alpha = b + 1$ and $\beta = b - 1$.Subsequently, the problem would be to maximize $\frac{2}{||\boldsymbol{w}||}$ or minimize $\frac{||\boldsymbol{w}||}{2}$.
- To make the problem easier when taking the gradients, we'll, instead, word with its squared form:

$$\min_{\boldsymbol{w},b} \quad \frac{1}{2}||\boldsymbol{w}||^2 \equiv \min_{\boldsymbol{w},b} \quad \frac{1}{2}\boldsymbol{w}^T\boldsymbol{w}$$

- This optimization comes with some constraints.
- Let's assume that the labels for our classes are {-1, +1}.
- When classifying the data points, we want the points belonging to positives classes to be greater than +1, meaning $\boldsymbol{w}^T\boldsymbol{x}+b \geq 1$ and the points belonging to the negative classes to be less than -1, i.e. . $\boldsymbol{w}^T\boldsymbol{x}+b \leq -1$
- We can combine these two constraints and express them as: $y_i(\boldsymbol{w}^T\boldsymbol{x}_i+b) \geq 1$
  Therefore our optimization problem would become: $\min_{\boldsymbol{w},b} \frac{1}{2}\boldsymbol{w}^T\boldsymbol{w}$ i.e $\textbf{s.t.} \quad y_i(\boldsymbol{w}^T\boldsymbol{x}_i+b) \geq 1$

- This optimization is called the primal problem and is guaranteed to have a global minimum.
- We can solve this by introducing Lagrange multipliers (α_i) and converting it to the dual problem:

$$L(\boldsymbol{w}, b, \alpha_i) = \frac{1}{2}\boldsymbol{w}^{\boldsymbol{T}}\boldsymbol{w} - \sum_{i=1}^{n} \alpha_i (y_i(\boldsymbol{w}^{\boldsymbol{T}}\boldsymbol{x}_i + b) - 1)$$

SVM with Soft Margin

- Misclassifications may happen.
- First,need to minimize the misclassification error, which means that we'll have to deal with one more constraint.
- Second, to minimize the error, we should define a loss function.
- A common loss function used for soft margin is the hinge loss.

$$\max\{0, 1 - y_i(\boldsymbol{w}^T \boldsymbol{x}_i + b)\}$$

- The loss of a misclassified point is called a slack variable and is added to the primal problem that we had for hard margin SVM. So the primal problem for the soft margin becomes:

$$\min \frac{1}{2}||\boldsymbol{w}||^2 + C\sum_{i=1}^{n} \zeta_i$$

$$\text{s.t. } y_i(\boldsymbol{w}^T\boldsymbol{x}_i + b) \geq 1 - \zeta_i \quad \forall i = 1, \ldots, n, \ \zeta_i \geq 0$$

- A new regularization parameter controls the trade-off between maximizing the margin and minimizing the loss.
- The difference between the primal problem and the one for the hard margin is the addition of slack variables.
- The new slack variables add flexibility for misclassifications of the model

Dual Problem

$$\max \; -\frac{1}{2}\sum_{i=1}^{n}\sum_{j=1}^{n}\alpha_i\alpha_j y_i y_j x_i^T x_j + \sum_{i=1}^{n}\alpha_i$$

$$\textbf{s.t.} \quad \sum_{i=1}^{n}\alpha_i y_i = 0, \;\; 0 \le \alpha_i \le C$$